Angelos Poulis

Boston, MA, USA

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Education

Boston University (BU) *Boston, MA*

Ph.D. in Computer Science *Sep 2024 – Present*

- *Advisors*: Evimaria Terzi, Mark Crovella
- Research interests: Interpretability of large language models
- Anticipated Graduation: May 2029
- Relevant graduate coursework: Machine Learning, Natural Language Processing, Interpretable Machine Learning, Algorithmic Data Mining

National and Kapodistrian University of Athens (NKUA) *Athens, Greece*

B.Sc. in Informatics and Telecommunications *Sep 2019 – Mar 2024*

- Major: Computer Science, Specialization: Data and Knowledge Management
- GPA: 8.7/10 (Honors)
- *Thesis*: "Reasoning over Description Logic-based Contexts with Transformers"
- *Supervisor*: Manolis Koubarakis, Co-supervisor: Eleni Tsalapati

Research Experience

Boston University *Boston, MA*

Graduate Research Assistant *Sep 2024 – Present*

- Studying the internal representations of large language models for interpretability, with a focus on probing high-level features and evaluating their generalization across tasks and settings.
- *Advisors:* Evimaria Terzi and Mark Crovella.

National and Kapodistrian University of Athens *Athens, Greece*

Research Intern, KDD lab *Oct 2023 – Jul 2024*

- Contributed to the AutoFair project, developing an automated toolkit for fairness evaluation of AI/ML models.
- Implemented components of an end-to-end pipeline for bias detection, fairness trade-off analysis, certification, and model explainability.
- *Supervisor:* Dimitrios Gunopulos.

National and Kapodistrian University of Athens *Athens, Greece*

Undergraduate Research Student, AI Team *Jan 2022 – Feb 2024*

- Created DELTA$_D$, a 384K-example synthetic benchmark for evaluating natural-language reasoning over Description Logic-based contexts.
- Fine-tuned transformer language models (RoBERTa and DeBERTa), and evaluated GPT models in zero- and few-shot settings.
- Analyzed model generalization across reasoning depth, linguistic complexity, and symbolic variations.
- *Supervisor:* Manolis Koubarakis.

Publications

A. Poulis, M. Crovella, E. Terzi. "Testing the Limits of Truth Directions in LLMs." To appear in the *Proceedings of the 9th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, 2026. arXiv:2604.03754.

A. Poulis, E. Tsalapati, M. Koubarakis. "Transformers in the Service of Description Logic–Based Contexts." In *Proceedings of EKAW 2024*, LNAI/LNCS, Springer, 2025. Earlier version presented at the NeLaMKRR Workshop at KR 2024.

Research Presentations

A. Poulis, M. Crovella, E. Terzi. "Testing the Limits of Truth Directions in LLMs." *NEMI Workshop*, 2026.

A. Poulis, M. Crovella, E. Terzi. "Using Geometries of Truth to Identify Factual Recall in LLMs." *NEMI Workshop*, 2025.

A. Poulis, D. Tomaras, V. Kalogeraki, D. Gunopulos. "AutoFair Toolkit: An Automated Tool for Evaluating Fairness in AI/ML Models." *EuroSys Posters Track*, 2024.

Skills / Technologies

Python • C/C++ • Java • SQL • PyTorch • Hugging Face Transformers • TransformerLens • nnsight • scikit-learn • NumPy

Honors and Awards

2025 **Scholarship**, Gerondelis Foundation

2025 **Scholarship**, Eleni Gatzoyiannis Foundation

Spring 2025 **Dean's Fellowship**, Department of Computer Science, BU

Fall 2024 **Non-service Chair's Award**, Department of Computer Science, BU

2024 **Best Thesis Award**, Department of Informatics and Telecoms., NKUA

2018, 2019 **Cyprus Olympiad in Informatics Finalist**, Cyprus Computer Society

Teaching Experience

Spring 2026 **BU CS132: Geometric Algorithms (Linear Algebra)**, Graduate Teaching Fellow — *Boston, MA*

Fall 2023 **NKUA K22: Operating Systems**, Teaching Assistant — *Athens, Greece*

Outreach

2020–Present **Cyprus Olympiad in Informatics**, Scientific Committee Member

Nov 24–29, 2021 **Cyprus Olympiad in Informatics**, Team Deputy Leader